POWER MARKETS

Casebook Q&A Overview: Applied Decision Cases

The casebook turns the lectures into bounded decisions: quote a product, size storage, screen a site, audit a partner, and choose an operating model under incomplete information.

What is a case for?

QUESTION What makes the casebook different from another summary chapter?

ANSWER Each case starts from a commercial situation and ends in a concrete artifact: a quote stack, duration hurdle, site scorecard, revenue-quality ladder, partner scorecard, product screen, or capability roadmap.

DEEPEN IN Cases C01-C07; Vol II Ch 7, Ch 16; Vol III Ch 12.

1

Situation

- A site, customer, battery, partner, portfolio, or operating-model question has incomplete data and several plausible paths.

2

Work product

- The answer has to be written as something a team could review, challenge, and improve.

3

Stress test

- One assumption changes after the recommendation.
- The case is useful when the change exposes the hinge variable.

4

Revision

- The artifact improves by changing evidence, assumptions, constraints, and decision boundaries.

What decision does each case stress?

QUESTION How do the seven cases cover the applied layer of the course?

ANSWER The cases cover recurring renewable-IPP decisions where physical limits, market structure, contract promises, and governance meet.

DEEPEN IN Cases C01-C07.

Case map

Case	Decision object	Primary artifact	Main constraint
C01	Shaped solar+BESS PPA	Quote stack	Customer promise versus residual exposure
C02	1h / 2h / 4h duration	Duration hurdle	Marginal MWh value versus constraints
C03	Grid-constrained site	Site scorecard	Grid rights can dominate resource quality
C04	BESS stack under saturation	Revenue-quality ladder	One asset spends the same MW, MWh, SoC, and cycle budget
C05	BRP/BSP performance	Partner scorecard	Execution needs a feasible benchmark
C06	Data-centre clean-power product	Product boundary screen	Hourly residual risk and evidence
C07	Year-one operating model	Capability roadmap	Build only what the portfolio can govern

C01: how is a shaped solar+BESS PPA quoted?

QUESTION Where is the offer boundary when a customer asks for smoother clean supply than raw solar?

ANSWER The seller has to price the transformation from production shape to customer promise and refuse residual risks that cannot be bounded or controlled.

RELATION Quote stack = raw energy + shaping/firming + imbalance/grid + evidence/credit + residual-risk margin.

DEEPEN IN Case C01; Vol II Ch 4, Ch 5, Ch 6, Ch 7; Vol III Ch 5.

1 Define the promise

- Shape, settlement basis, certificates, data obligations, tolerance bands, and fallback rules.

2 Estimate residual exposure

- Solar, BESS, portfolio support, and market procurement reduce hard-hour risk while leaving a residual tail.

3 Price the stack

- Energy, shape, imbalance, grid, evidence, credit, optionality, and margin.

4 Set the boundary

- The offer breaks when residual risk, control rights, or fallback obligations become too open-ended for the price.

C02: when is extra storage duration justified?

QUESTION How are 1h, 2h, and 4h BESS options compared?

ANSWER Extra duration is a marginal investment with its own scarce hours, capex, degradation, warranty, insurance, tariff, and product-depth test.

RELATION Required energy = MW obligation × duration, but the investment case depends on marginal net revenue and constraints.

DEEPEN IN Case C02; Vol I Ch 13; Vol II Ch 9, Ch 12; Vol III Ch 6.

1 Find the marginal scarcity

- Identify which hours require additional MWh after shorter duration is exhausted or reserved.

2 Price the incremental asset

- Include capex, augmentation, degradation, grid impact, warranty and insurance restrictions, and opportunity cost.

3 Test market depth

- Competition and rule changes can make a real product too shallow for the extra capacity.

4 Decide

- Extra duration is attractive only when the marginal MWh clears its own hurdle under realistic operating limits.

C03: which grid-constrained site advances?

QUESTION How can a lower-resource site outrank a higher-resource site?

ANSWER The investable site is the one whose resource, connection, curtailment, tariff, and optionality package creates the better risk-adjusted path.

RELATION Site value changes with delivered MWh, curtailment, connection timing, tariff structure, and flexibility rights.

DEEPEN IN Case C03; Vol I Ch 4; Vol II Ch 2; Vol III Ch 3.

1

Separate resource from grid right

- Irradiation or wind quality is only one input.

2

Score the wire side

- Export/import capacity, queue risk, curtailment, connection cost, tariffs, losses, and local flexibility.

3

Test co-location

- Storage can improve or worsen the constraint depending on grid rights and operating mode.

4

Rank the sites

- A weaker resource with a better connection can beat a stronger resource behind costly or uncertain access.

C04: how robust is a BESS stack under saturation?

QUESTION What happens when many batteries chase similar revenue products?

ANSWER The stack has to be tested for mutually exclusive use of MW, MWh, SoC, cycle budget, grid rights, qualification, and control rights.

ANCHOR The same MW, MWh, state-of-charge headroom, and cycle budget cannot be spent twice.

DEEPEN IN Case C04; Vol II Ch 8, Ch 10, Ch 12; Vol III Ch 1.

1 Candidate revenues

- Arbitrage, reserve capacity, balancing energy, co-location, grid service, customer product, and optionality.

2 Scarce-resource map

- Each line claims physical capability or operating rights from the same asset.

3 Revenue quality

- Separate underwritten floor, repeatable market value, optimizer forecast, merchant upside, and option value.

4 Decision hinge

- A revenue line can be real and still too small, too crowded, or too operationally expensive to carry the investment case.

C05: how is BRP/BSP performance audited?

QUESTION How can an asset owner judge outsourced execution without hindsight bias?

ANSWER The audit needs raw data and a feasible counterfactual built from the information, rights, liquidity, and constraints available at the time.

DEEPEN IN Case C05; Vol I Ch 8, Ch 9; Vol II Ch 15; Vol III Ch 10.

1 **Collect evidence**

- Forecasts, bids, schedules, activations, metering, settlement, constraints, timestamps, and contract rights.

2 **Build the benchmark**

- Use only actions that were feasible at the relevant decision time.

3 **Attribute leakage**

- Market conditions, forecast error, poor bidding, conservative limits, data problems, and contract design have different remedies.

4 **Escalate**

- Escalation is credible when materiality, feasibility, and contractual responsibility are visible.

C06: where is the boundary of a data-centre clean-power product?

QUESTION What has to be bounded before selling a 24/7-style or hourly clean-power claim?

ANSWER The seller has to define load shape, deliverable supply, firming, control rights, certificate evidence, residual hard hours, and fallback rules.

DEEPEN IN Case C06; Vol III Ch 5, Ch 7; Vol II Ch 4, Ch 6.

Product-boundary screen

Screen	Required answer	Failure mode
Load	Hourly demand and flexibility	Annual average hides hard hours
Supply	Eligible assets by zone and interval	Correlated or remote shortfall
Firming	Who covers residual hours	Uncapped market purchases
Control	Dispatch, curtailment, and data rights	Promise without operating rights
Evidence	Certificates, meters, audit trail	Claim quality cannot be defended

C07: what operating model belongs in year one?

QUESTION Which capabilities belong in build, buy, outsource, audit, or postpone first?

ANSWER Build around recurring decisions that carry material risk and require proprietary judgment; outsource execution only with data rights and benchmarks.

DEEPEN IN Case C07; Vol II Ch 16, Ch 17; Vol III Ch 10, Ch 11, Ch 12.

Capability roadmap

Decision area	Year-one posture	Reason
PPA quote stack	Build internal method	Pricing logic and risk boundary are strategic
BESS dispatch benchmark	Build/audit	Partner actions need feasible comparison
Market execution	Outsource / hybrid	24/7 operations need scale
Data contracts	Build internal control	Every later model depends on evidence
Risk reporting	Build lightweight pack	Management needs recurring decision visibility
Full trading desk	Postpone	Governance and volume threshold come later

How does a reader use a case?

QUESTION What working habit does the casebook train?

ANSWER Write a first decision, name the scarce variable, then compare it with the worked artifact and rerun the case after one constraint changes.

DEEPEN IN Cases C01-C07.

1

First answer

- State the decision in one sentence.
- Name the scarce variable: MW, MWh, connection, credit, liquidity, data rights, cycle budget, or residual-risk tolerance.

2

Worked artifact

- Compare the artifact with the first answer.
- Track which risk was assigned, priced, bounded, or left unaccepted.

3

Constraint flip

- Change one assumption and rerun the decision.
- The best cases show which variable actually controlled the answer.

What does the casebook stress?

QUESTION Which failure modes recur across the cases?

ANSWER The same themes repeat: product promises can exceed capability, physical resources cannot be spent twice, finance separates revenue quality, and governance needs evidence.

DEEPEN IN **Cases C01-C07; Vol II Ch 8-Ch 17; Vol III Ch 10-Ch 12.**

Commercial and physical stress

- Product promises can outrun asset capability, evidence, or control rights.
- MW, MWh, grid access, SoC, degradation, and response requirements cannot be spent twice.

Financial and governance stress

- Bankable revenue, merchant upside, credit exposure, and liquidity risk have different uses.
- Outsourced execution can work only when data rights, feasible benchmarks, and escalation rules exist.